Fulbright

Operating Systems

CS211 – Spring 2026

(December 29th, 2025)

1. Course information

1.1. Logistics

- Instructor: Linh Huynh (linh.huynh@fulbright.edu.vn)
- Lecture: Friday (1:15pm - 4:15pm), CR7
- Office Hours: After the lectures (4:15pm - 4:30pm) or Wed (1pm - 3pm)

1.2. Course Description

The operating system serves as an interface between user programs and hardware devices. Understanding its workings is essential for both application developers, as it aids in enhancing application performance, and systems developers, who may need to extend or create new operating systems for emerging devices. This course addresses this necessity by introducing fundamental concepts and principles of modern operating systems. The course is divided into three main parts. The first part delves into concurrency, covering topics such as processes, threads, CPU scheduling, context switching, synchronization, and deadlock. The second part explores memory management issues, including linking, allocation, address translation, segmentation, paging, and swapping. The third part introduces file systems, discussing disk management, file organization, and crash recovery. After these three parts, the course will conclude with advanced topics such as scalability, virtualization, security, and support for distributed computing.

1.3. Prerequisites:

1.3.1. CS201 - Data Structures & Algorithms

1.3.2. CS203 - Computer Organization.

1.4. Course's learning outcomes

Students who pass this course will achieve the following course learning outcomes (CLO)

(CLO1) be able to define the key components and terminologies in operating systems.

(CLO2) be able to explain the mechanisms used in different operating system components such as cpu scheduling, memory management and input/output device management.

(CLO3) be able to analyze the advantages/disadvantages of algorithms related to cpu scheduling, process synchronization, and memory management.
(CLO4) be able to compare the multi-process and multi-thread implementation of an application.
(CLO5) be able to analyze the effect of virtual memory management on program performance.
(CLO6) be able to analyze and develop simple system-level programs.
(CLO7) be able to work in a group to study features of a new operating system

1.5. Assessment of Learning

Homework (7 problem sets):	35%
Two midterm exams (45-minute, writing):	30%
Final Exam (75-minute, writing):	25%
Project:	10%

1.6. Missed Exams & Late Homework

- Late homework is not accepted
- A makeup exam is only provided for excused absences (sickness, family events, or FUV events) and is harder than the original one.

1.7. Letter grade scale

Grade points are assigned to letter grades according to the following scale (unless otherwise specified by faculty in individual courses):

Letter Grade	Grade Point	Percentage
A	4.0	94-100%
A-	3.7	90-93%
B+	3.3	87-89%
B	3.0	83-86%
B-	2.7	80-82%
C+	2.3	77-79%
C	2.0	73-76%
C-	1.7	70-72%
D+	1.3	67-69%
D	1.0	60-66%
F	0.0	0-59%

1.8. P/NP (Pass/No Pass)

Please refer to the latest version of P/NP policy in the Grading Guide.

1.9. Required Materials

Textbooks (optional):

- Operating System Concepts (10th Edition) by Abraham Silberschatz, Peter B. Galvin, and Greg Gagne.
- Operating Systems: Principles and Practice (2nd Edition) by Thomas Anderson and Michael Dahlin.

- Operating Systems: Three Easy Pieces by Remzi H Arpaci-Dusseau and Andrea C Arpaci-Dusseau

1.10. Tentative Course Schedule

Week	Activity	Topics
1	Lecture 1 Lecture 2	Introduction Computer architecture review & systems programming
2-3	Lecture 3	Processes
4-5	Lecture 4	Concurrency & threads
6-7	Lecture 5 **Mid-term I**	Synchronization
8	Lecture 6	Scheduling
9-10	Lecture 7	Memory management
11	Lecture 8 **Mid-term II**	I/O, disks, and RAID
12	Lecture 9	File systems
13	Lecture 10	Crash recovery
14	Lecture 11	Special topics
15	**Final exam & Student presentation**	

2. Professional expectations & supports

2.1. Attendance

Fulbright students are expected to fully engage with the courses in which they are enrolled at Fulbright. Students are generally expected to miss no more than three days of class.

2.2. Time expectation

Students should spend at least 10 hours per week (including 3 hours in class) for this course.

2.3. Academic Dishonesty

As Fulbright University's **Code of Academic Integrity** explains: "*plagiarism occurs when a writer appropriates another's ideas, research, or writing without proper acknowledgement of the source or uses another's words without the use of*

quotation marks, whether intentional or not." All Fulbright students are responsible for familiarizing themselves with the Code of Academic Integrity. In order to help students become familiar with the concepts of academic integrity, citation, and plagiarism, all students are required to complete Level 1 and Level 2 of the Academic Integrity Credential (available on Canvas). Academic misconduct (including plagiarism, cheating, and other violations of the Code of Academic Integrity) will be investigated according to this code and may be subject to sanctions including reduced grade, failure, academic warning, academic probation, academic suspension, or withholding of degree. If you have questions about plagiarism, you should contact your teacher or the Learning Support Team at learningsupport@fulbright.edu.vn .

2.4. Accessibility Learning Service

Fulbright University Vietnam commits to providing excellent student-centered services that supports diversity, inclusivity and accessibility where the student's voice and presence matters. Accessibility Learning Service provides support for students with conditions, including disability, long-term illness, mental health condition or being primary carers of individuals with a disability. ALS can meet with you to develop individualized learning plan, share your plan with your professors and provide continuing support if necessary. You can contact us at wellness@fulbright.edu.vn to book an appointment. We strongly recommend that you meet with us prior to the semester start to ensure timely development and implementation of your learning plan.

2.5. Learning Support

If you have further questions about learning support, please send an email to learning.support@fulbright.edu.vn